Fuzzy Logic & Neural Networks (CS-514)

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Membership Functions

- Membership functions were first introduced in 1965 by Lofti A. Zadeh in his first research paper "fuzzy sets".
- Membership functions characterize fuzziness (i.e., all the information in fuzzy set), whether the elements in fuzzy sets are discrete or continuous.
- Membership functions can be defined as a technique to solve practical problems by experience rather than knowledge.
- Membership functions are represented by graphical forms.

Membership Functions

- In the fuzzy theory, fuzzy set A of universe X is defined by function $\mu_A(x)$ called the membership function of set A.

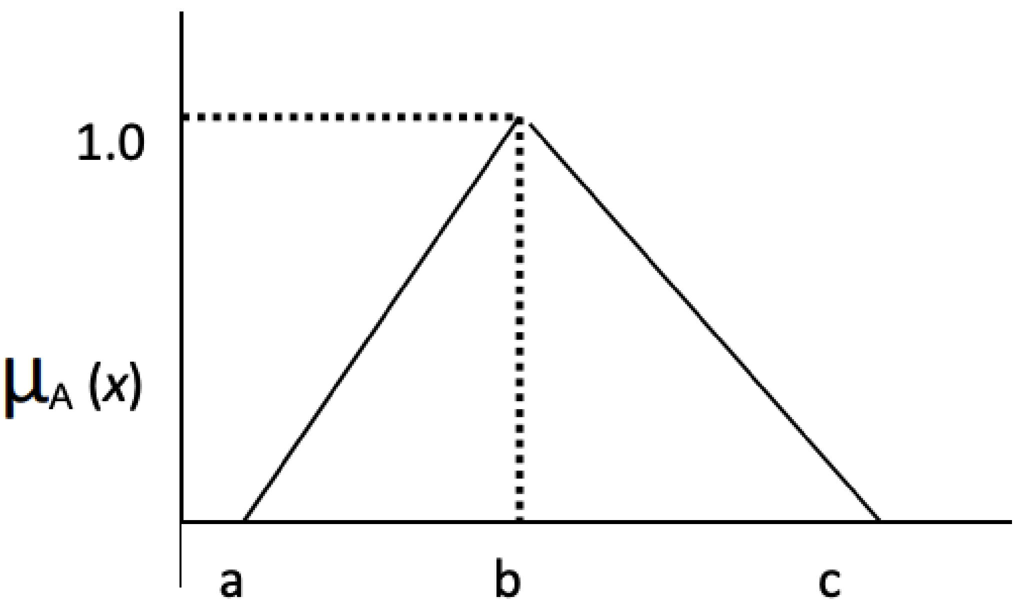
1) Triangular Membership
2) Trapezoidal MF
3) Gaussian MF
4) Generalized bell MF
5) Sigmoid membership function

Membership Functions

Triangular Membership Function:

$$\mu(x) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

$$\mu_A(x) = \begin{cases} 0 & if\ x \le a \\ \dfrac{x-a}{b-a} & if\ a \le x \le b \\ \dfrac{c-x}{c-b} & if\ b \le x \le c \\ 0 & if\ x \ge c \end{cases}$$



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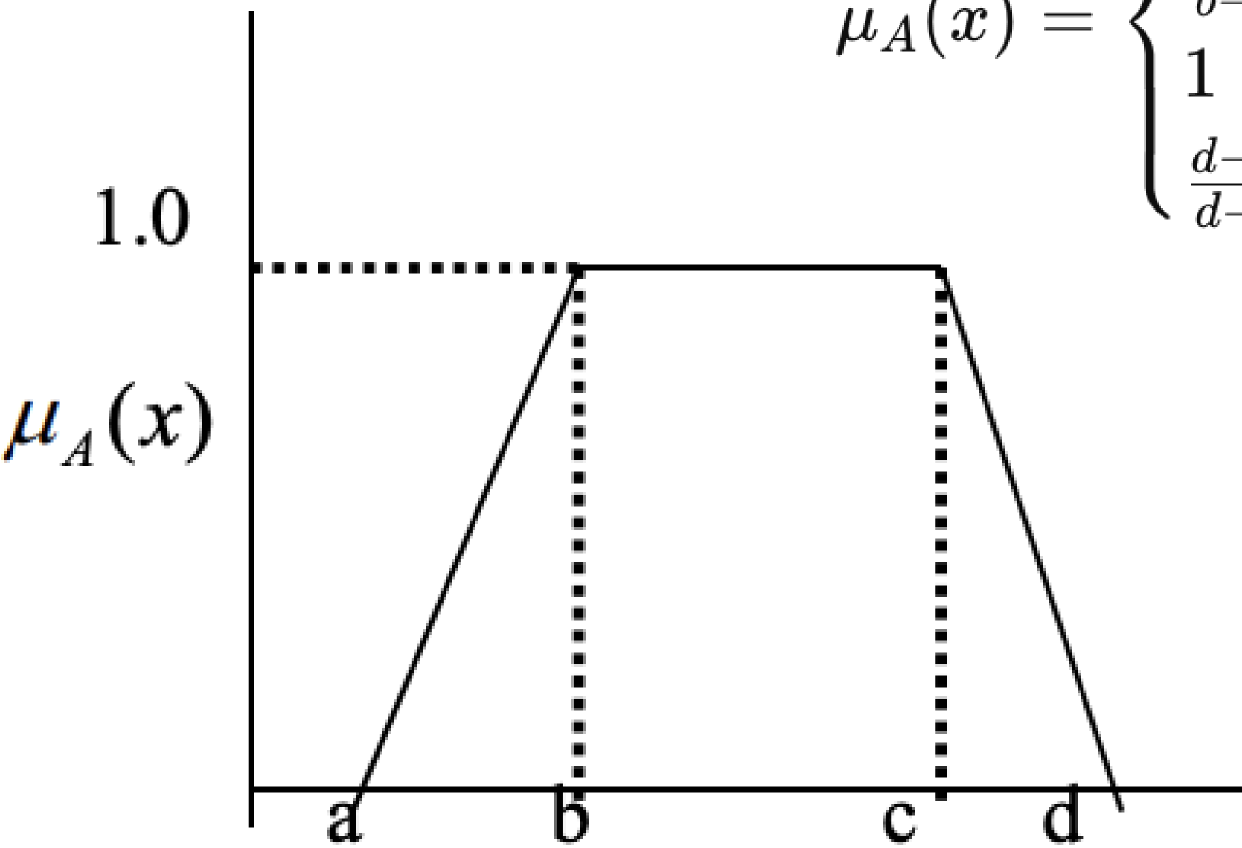
Membership Functions

Trapezoidal Membership Function:

$$\mu(x) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

$$\mu_A(x) = \begin{cases} 0 & x \le a \text{ or } x \ge d \\ \frac{x-a}{b-a} & a < x < b \\ 1 & b \le x \le c \\ \frac{d-x}{d-c} & c < x < d \end{cases}$$



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Membership Functions

Gaussian Membership Function:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$

where:

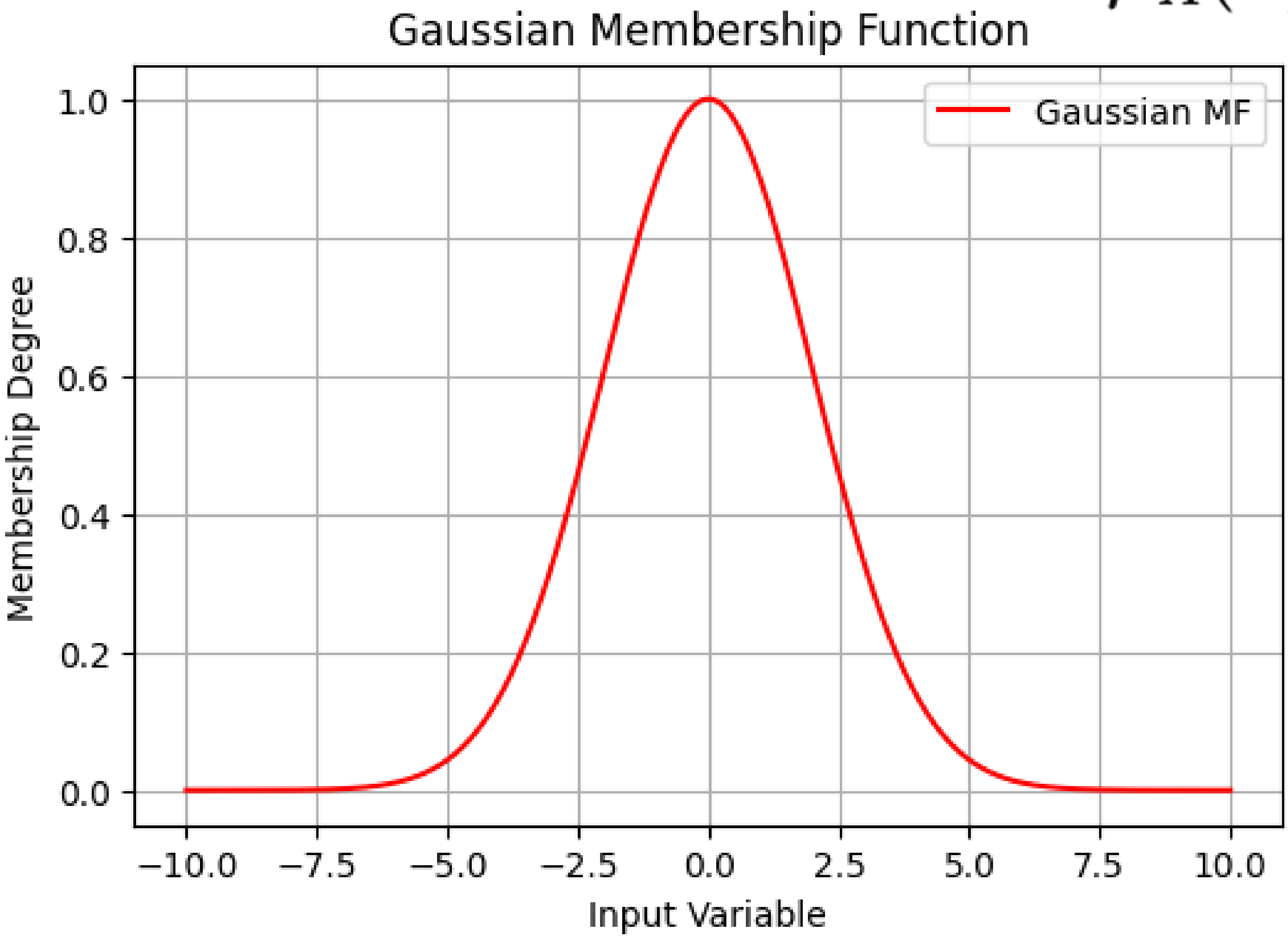
- c is the center of the curve (mean).
- σ is the standard deviation (spread of the curve).

Source: Internet

Membership Functions

Gaussian Membership Function:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$

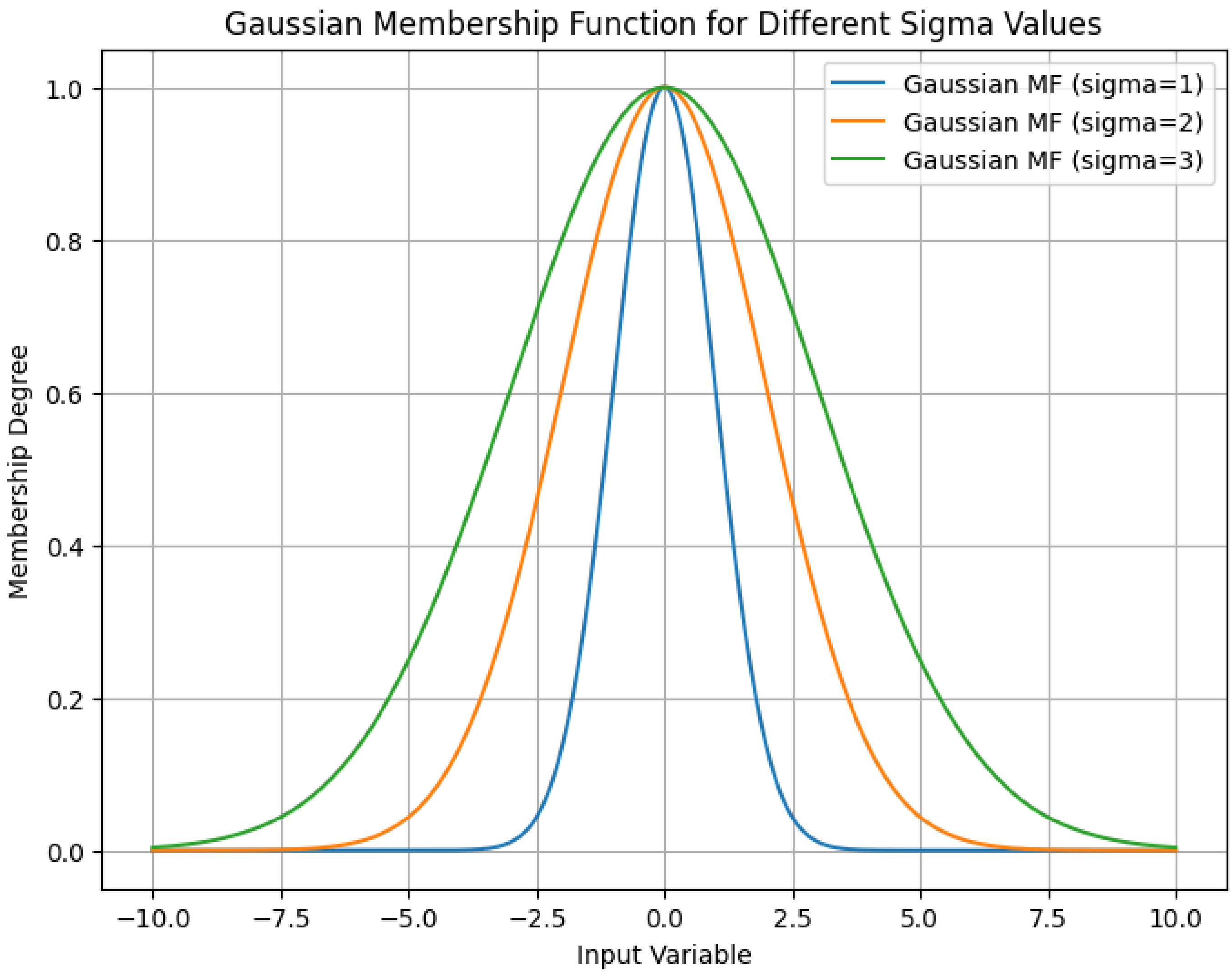


$$c = 0$$
$$\sigma = 3$$

Membership Functions

Gaussian Membership Function:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$



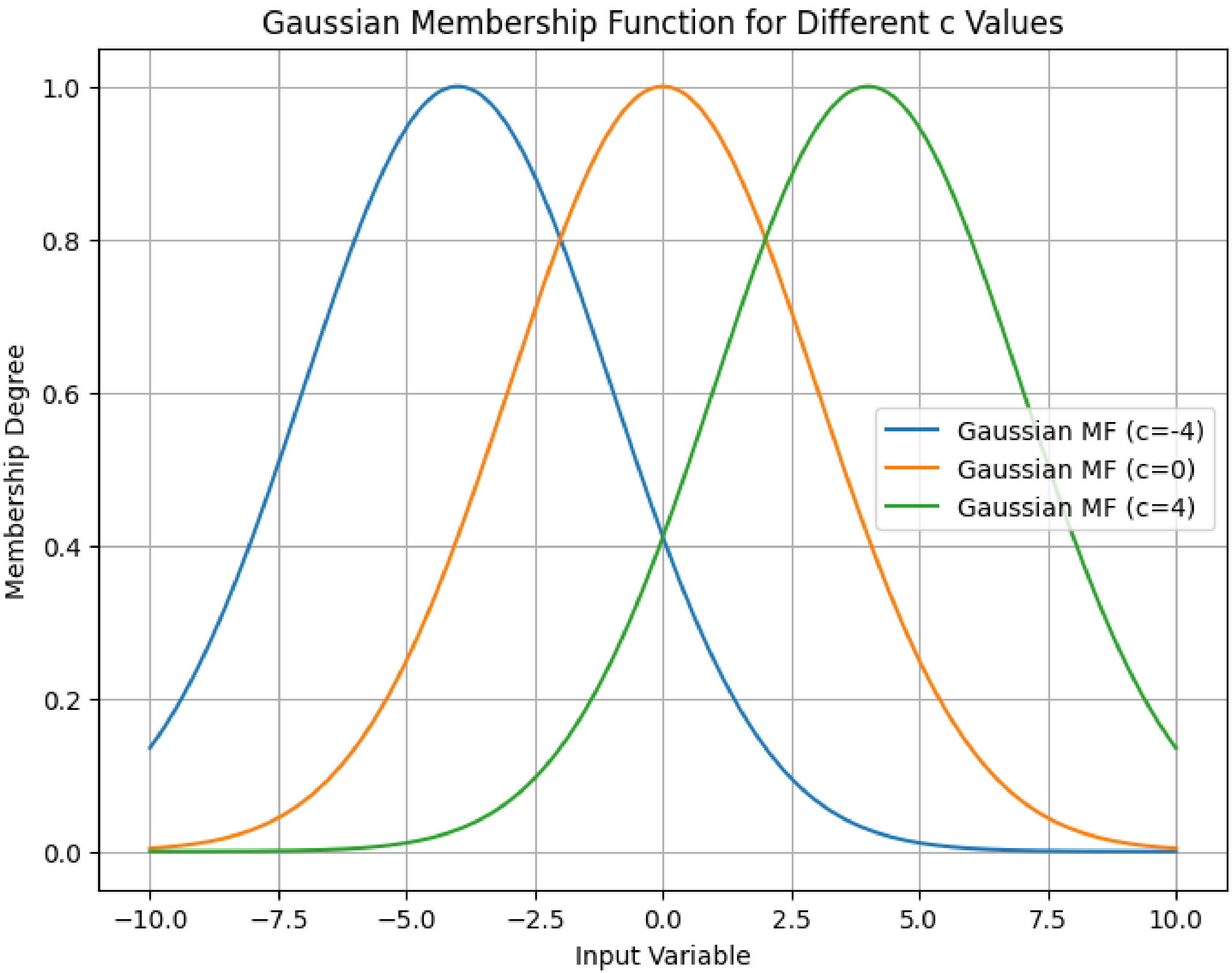
$$c = 0$$

$$\sigma = 1,2,3$$

Membership Functions

Gaussian Membership Function:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$



$$c = -4, 0, 4$$

$$\sigma = 3$$

Membership Functions

Generalized Bell Membership Function:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$

where:

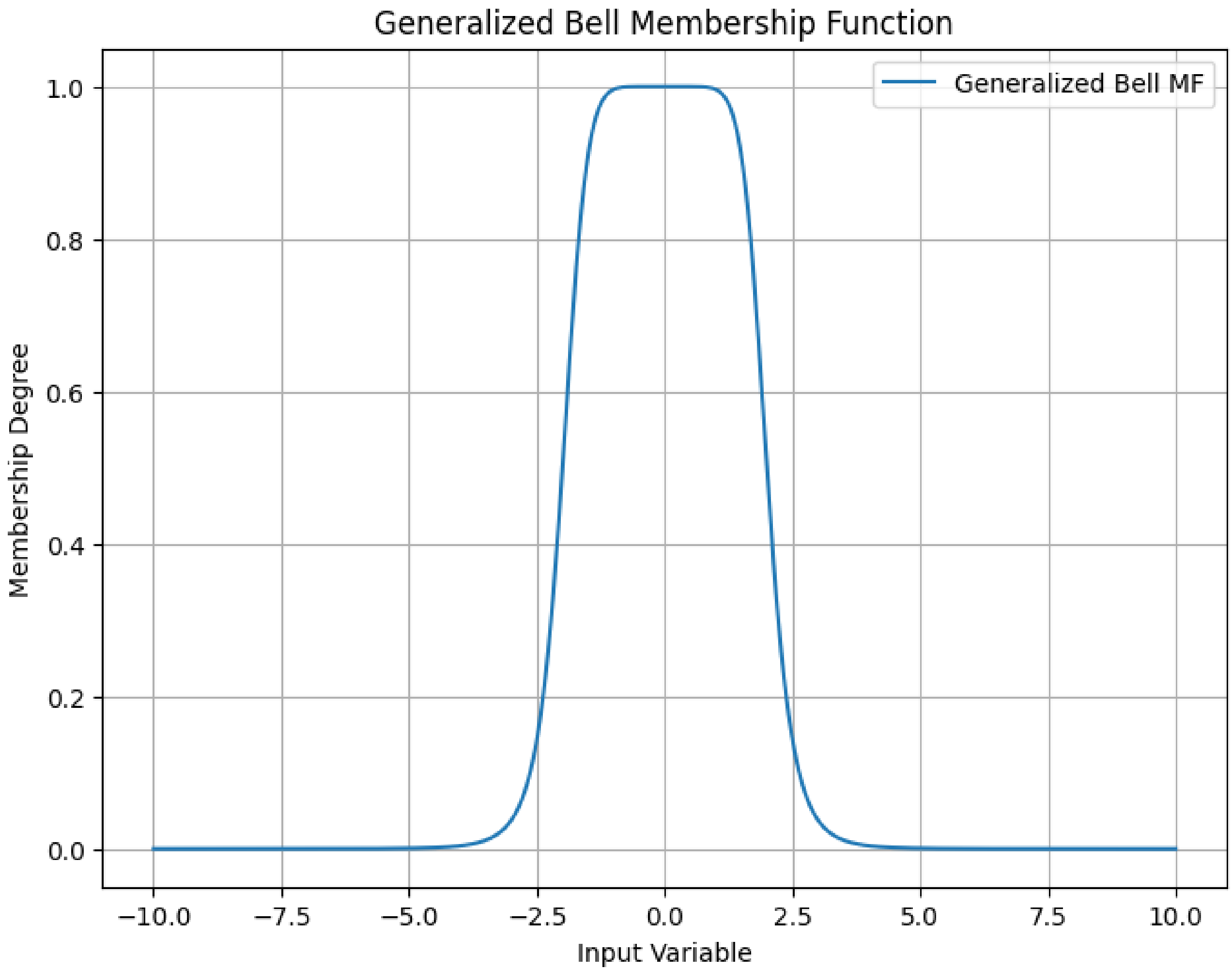
- a controls the width.
- b controls the slope.
- c is the center of the bell curve.

Source: Internet

Membership Functions

Generalized Bell Membership Function:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$

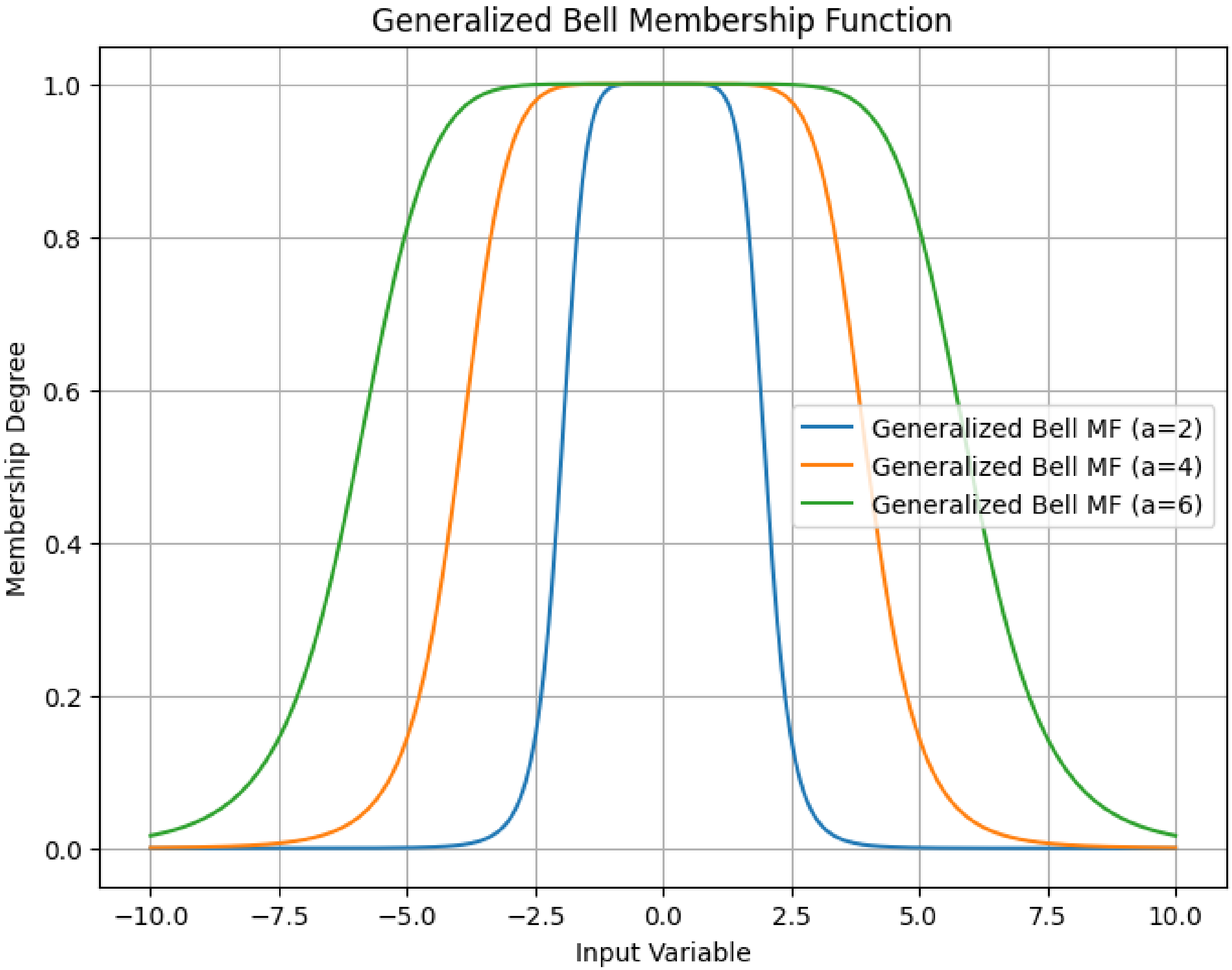


$$a = 2$$
$$b = 4$$
$$c = 0$$

Membership Functions

Generalized Bell Membership Function:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$



$a = 2{,}4{,}6$

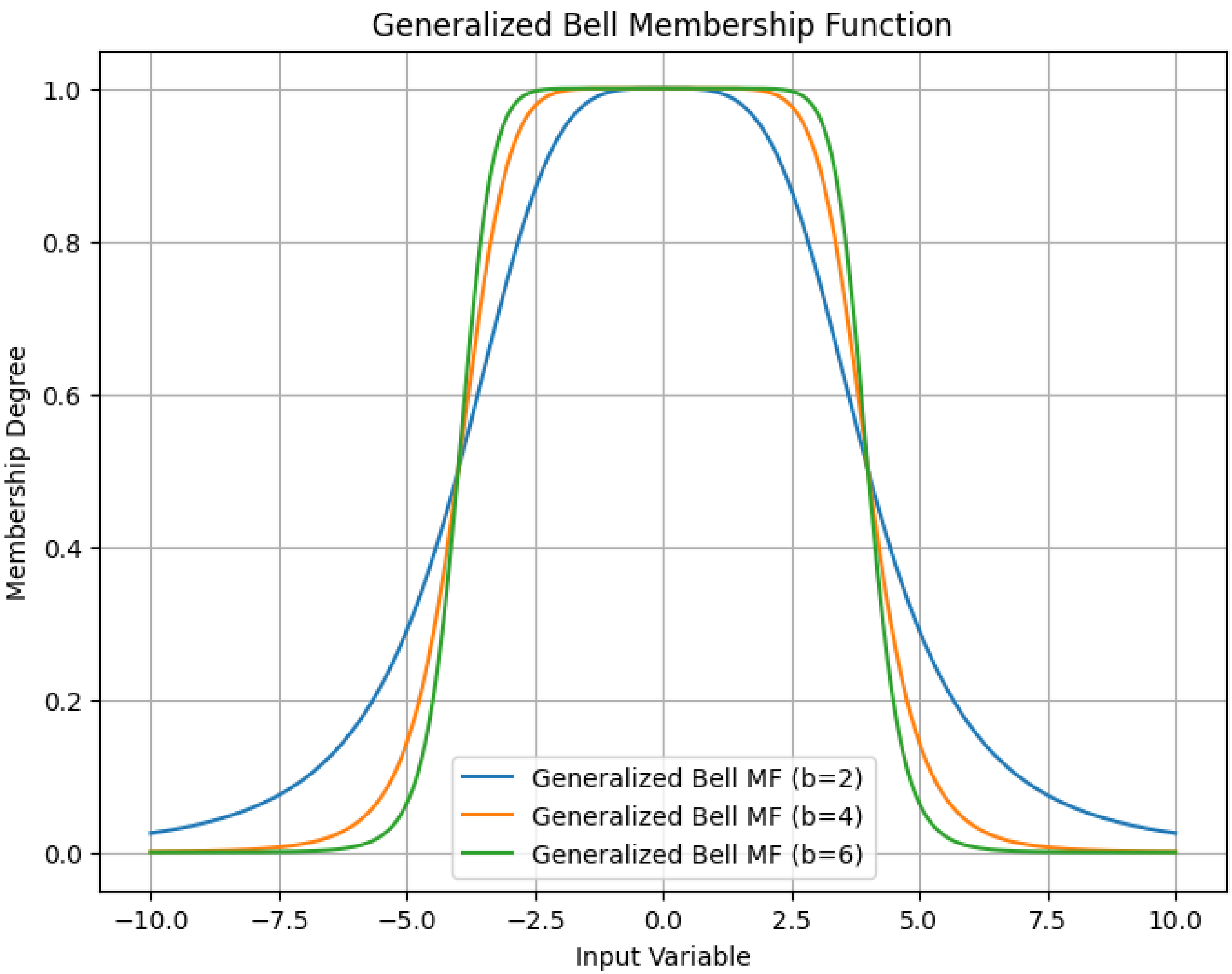
$b = 4$

$c = 0$

Membership Functions

Generalized Bell Membership Function:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$

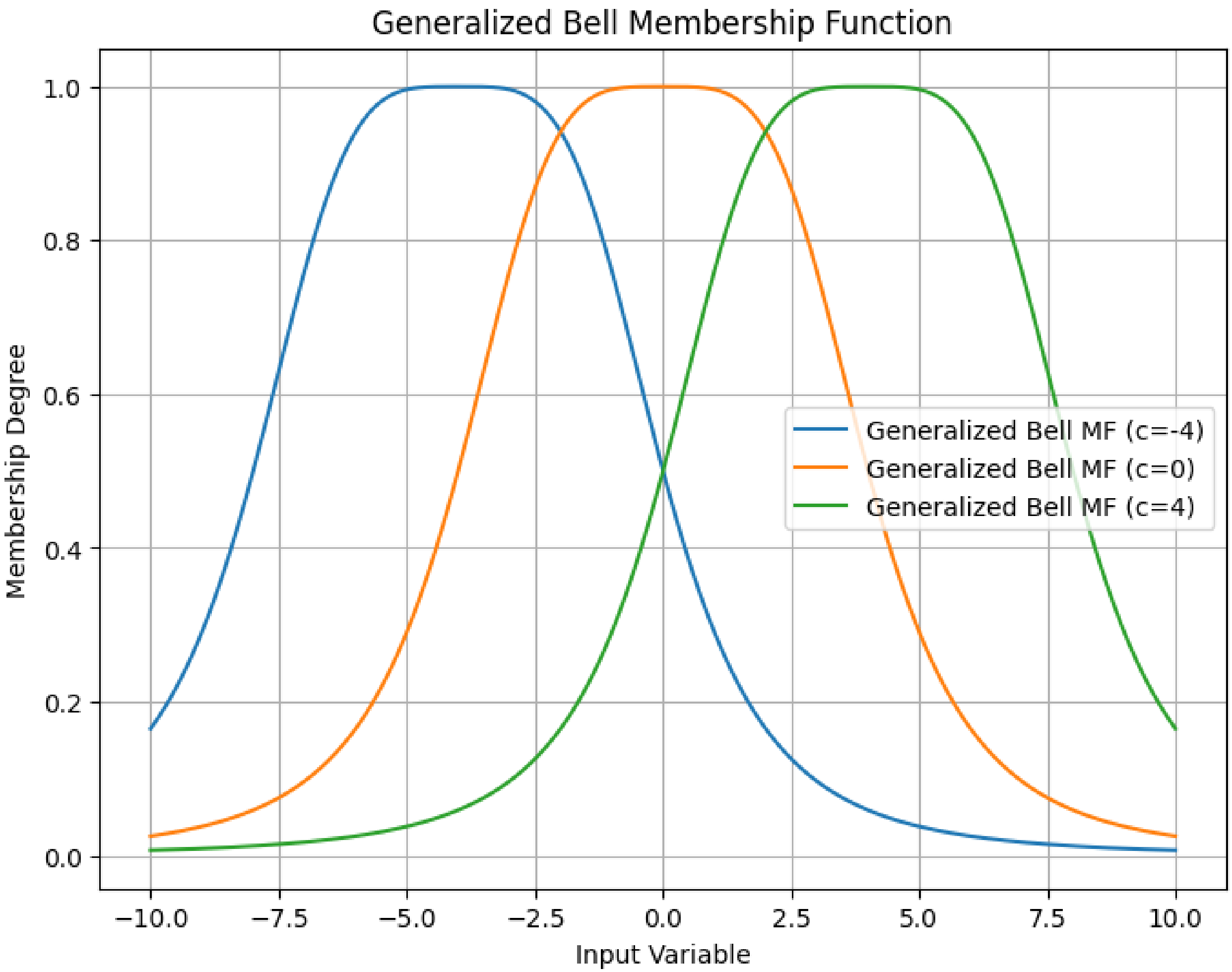


$$a = 2$$
$$b = 2,4,6$$
$$c = 0$$

Membership Functions

Generalized Bell Membership Function:

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$



$$a = 4$$
$$b = 2$$
$$c = -4,0,4$$

Membership Functions

Sigmoidal Membership Function:

$$\mu_A(x) = \frac{1}{1 + e^{-a(x-c)}}$$

where:

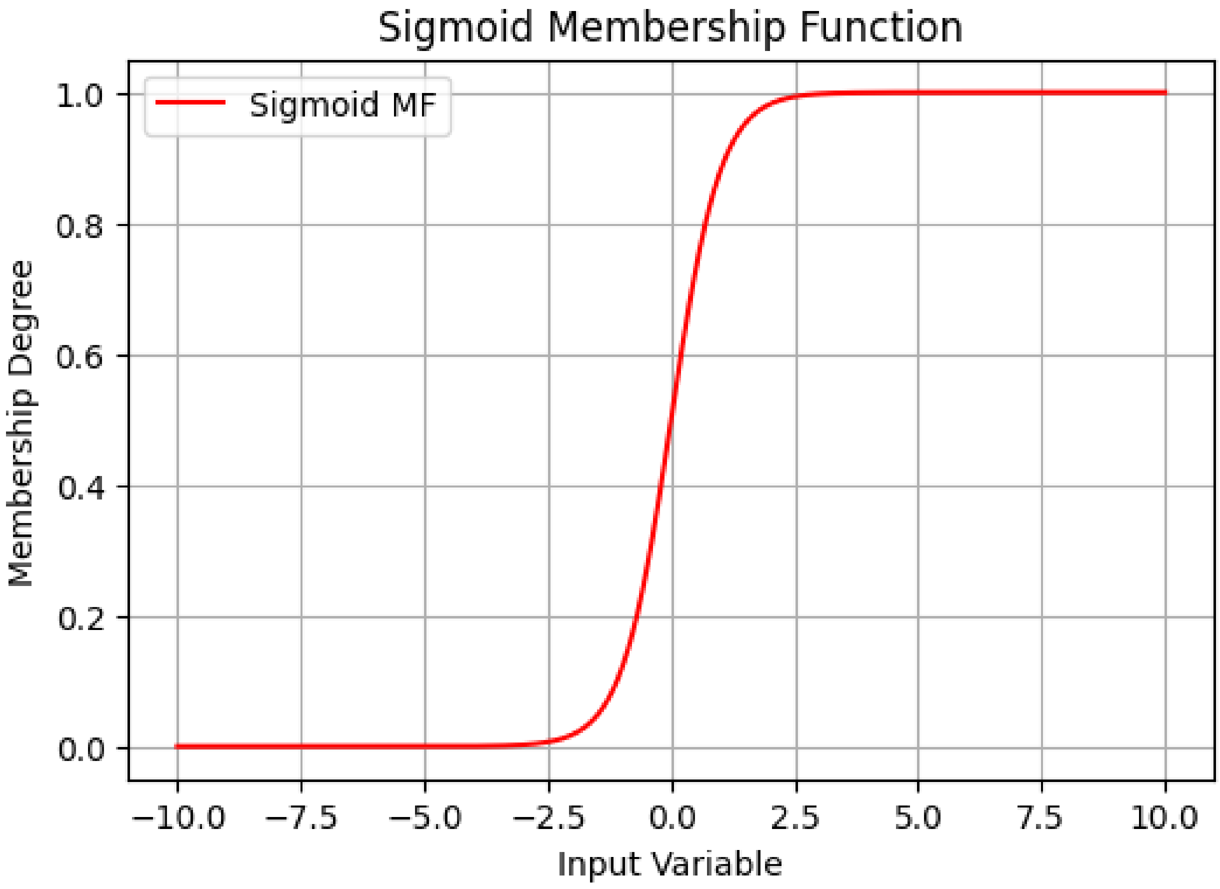
- a controls the slope of the curve.
- c is the center of the curve.

Source: Internet

Membership Functions

Sigmoidal Membership Function:

$$\mu_A(x) = \frac{1}{1 + e^{-a(x-c)}}$$



$$a = 2$$
$$c = 0$$

Membership Functions

Sigmoidal Membership Function:

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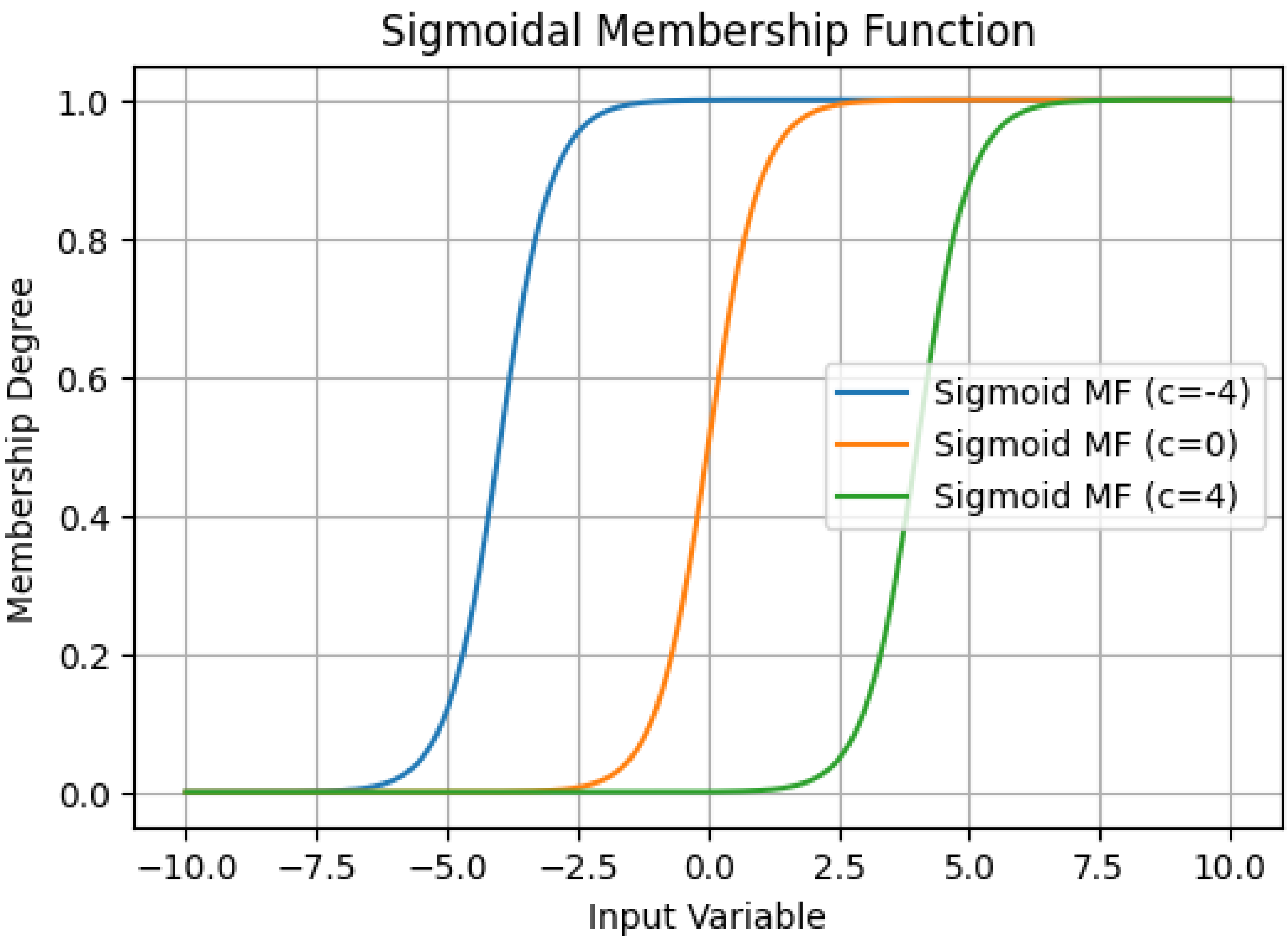

$$a = 2$$
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Membership Functions

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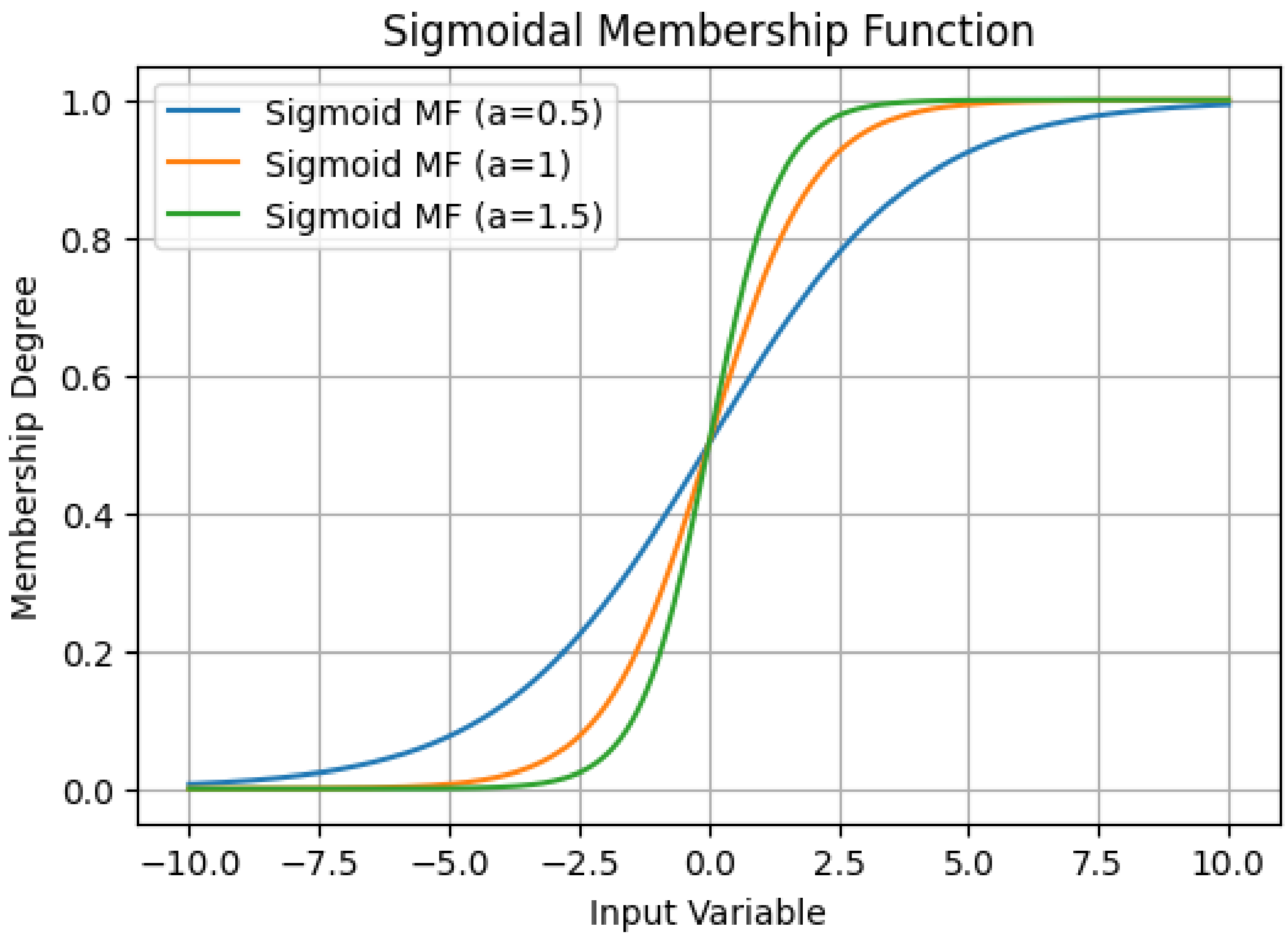
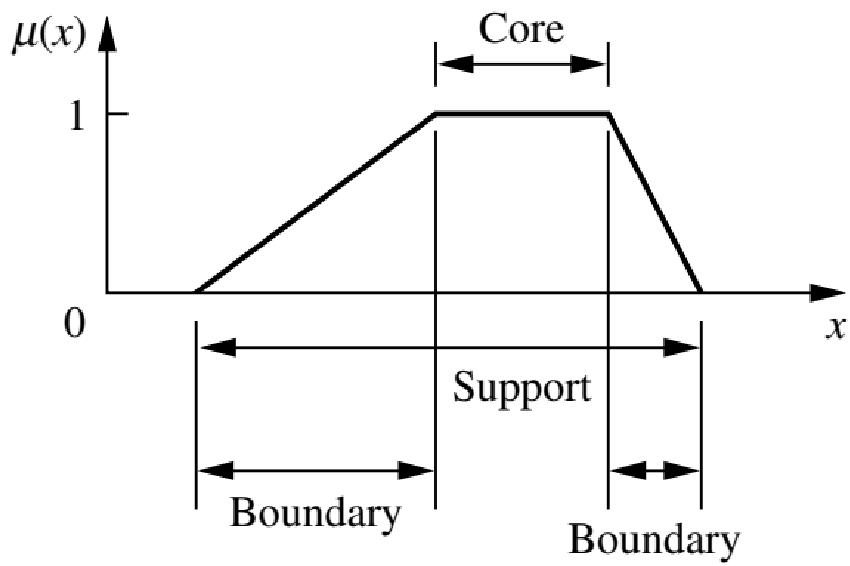


$$a = 0.5,1,1.5$$
$$c = 0$$

Membership Functions

Features of the Membership Function:

- The **core** of a membership function for some fuzzy set **A** is defined as $\mu_A(x) = 1$.
- The **support** of a membership function for some fuzzy set **A** is defined as $\mu_A(x) > 0$.
- The **boundaries** of a membership function for some fuzzy set **A** are defined as $0 < \mu_A(x) < 1$.



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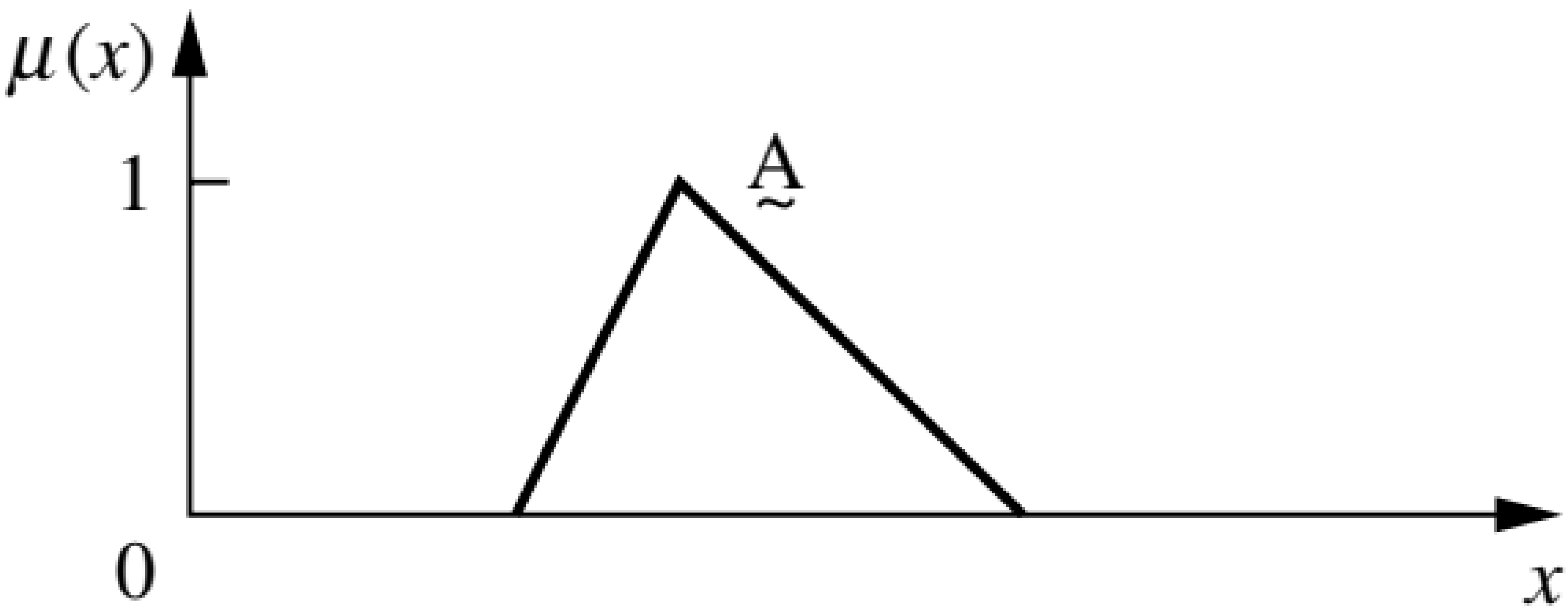
Membership Functions

Features of the Membership Function:

- A **normal** fuzzy set is one whose membership function has at least one element x in the universe whose membership value is unity.



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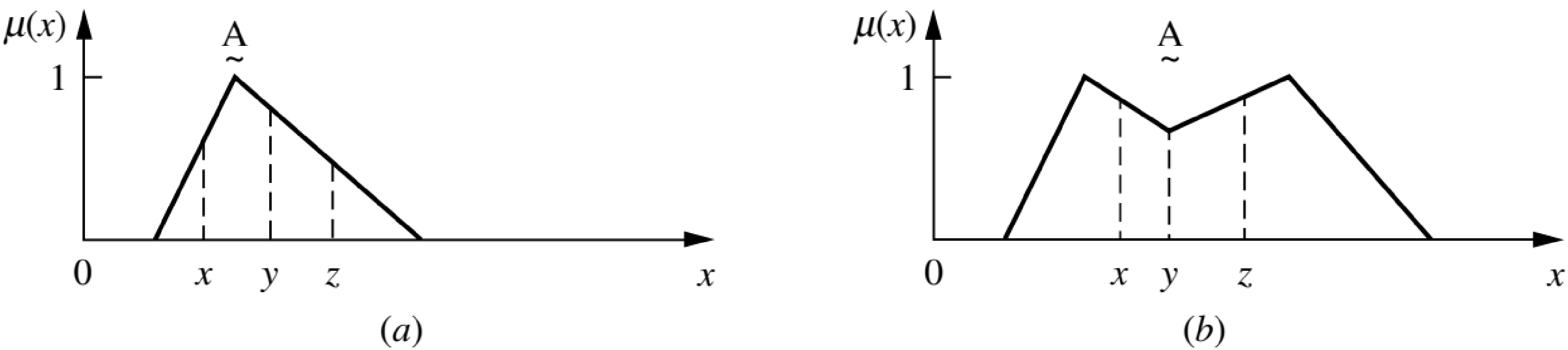
Membership Functions

Features of the Membership Function:

- A **convex** fuzzy set is described for any elements x, y, and z in a fuzzy set A, the relation $x < y < z$ implies that $\mu_A(y) \geq \min[\mu_A(x), \mu_A(z)]$



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Membership Functions

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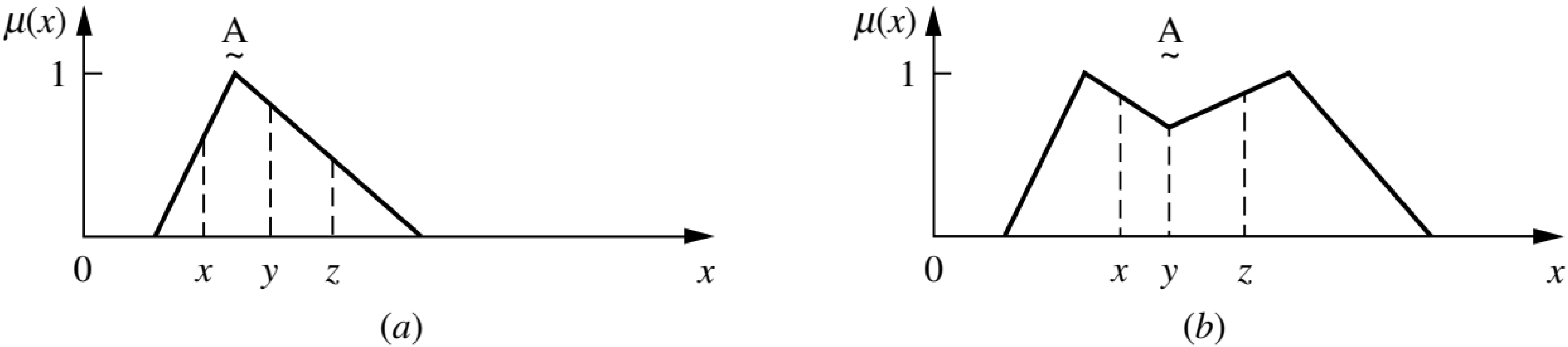

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Fuzzification

- In Fuzzification, the crisp (precise) inputs are converted into fuzzy sets using membership functions.
- There are several methods of fuzzification, each differs in how they assign degrees of membership to the input data.

The main methods are:

1. Singleton Fuzzification
2. Gaussian Fuzzification
3. Trapezoidal or Triangular Fuzzification
4. Piecewise Linear Fuzzification
5. Sigmoidal Fuzzification
6. Max-Min Fuzzification (Threshold-Based Fuzzification)
7. Discretization Method

Fuzzification

Singleton Fuzzification:

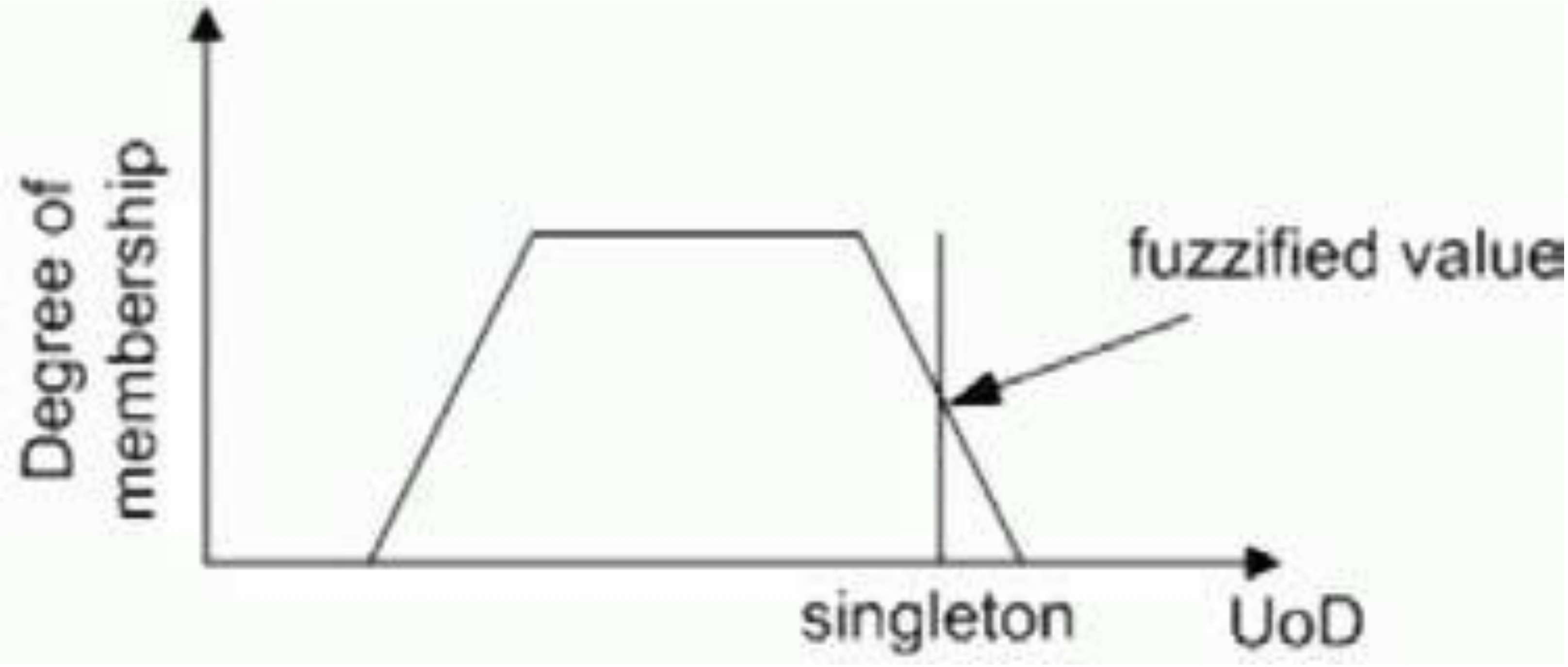
- In this method, the crisp input is associated with a specific degree of membership in only one fuzzy set.

- **Example:** If the temperature is 60°C, it may be assigned a degree of 1 (full membership) to the fuzzy set "Warm," without any membership in other sets like "Cold" or "Hot.”

- This method is typically used when the computation needs to be simple.

Fuzzification

Singleton Fuzzification:



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Fuzzification

Gaussian Fuzzification:

- This method uses a **Gaussian membership function** to assign degrees of membership to different fuzzy sets.

- **Example:** If temperature is 60°C, it might have a high membership in "Warm" and a lower membership in "Hot" or "Cold.”

- It allows for smoother transitions between fuzzy sets.

Fuzzification

Trapezoidal or Triangular Fuzzification:

- In this method, **triangular** or **trapezoidal membership functions** are used.

- **Example:** For the "Warm" fuzzy set a triangular membership of 1.0 occurs at 60°C, and gradually decreases as the temperature moves towards "Cold" or "Hot.”

- Easier to implement, computationally efficient.

Fuzzification

Piecewise Linear Fuzzification:

- This method uses **piecewise linear functions** to map crisp inputs into fuzzy sets.
- **Example:** In A fuzzy set for speed, the speed increases smoothly, but decreases sharply between "Slow" and "Medium".
- Flexible, allowing for custom shapes and transitions.

Fuzzification

Sigmoidal Fuzzification:

- In this method, a **sigmoid-shaped function** is used for fuzzification.
- **Example:** For temperature, a sigmoidal function might assign gradual membership to "Hot" or "Cold," with more rapid transitions as the temperature nears an extreme.
- Useful when transitions between states are asymmetric.

Fuzzification

Max-Min or Threshold-Based Fuzzification:

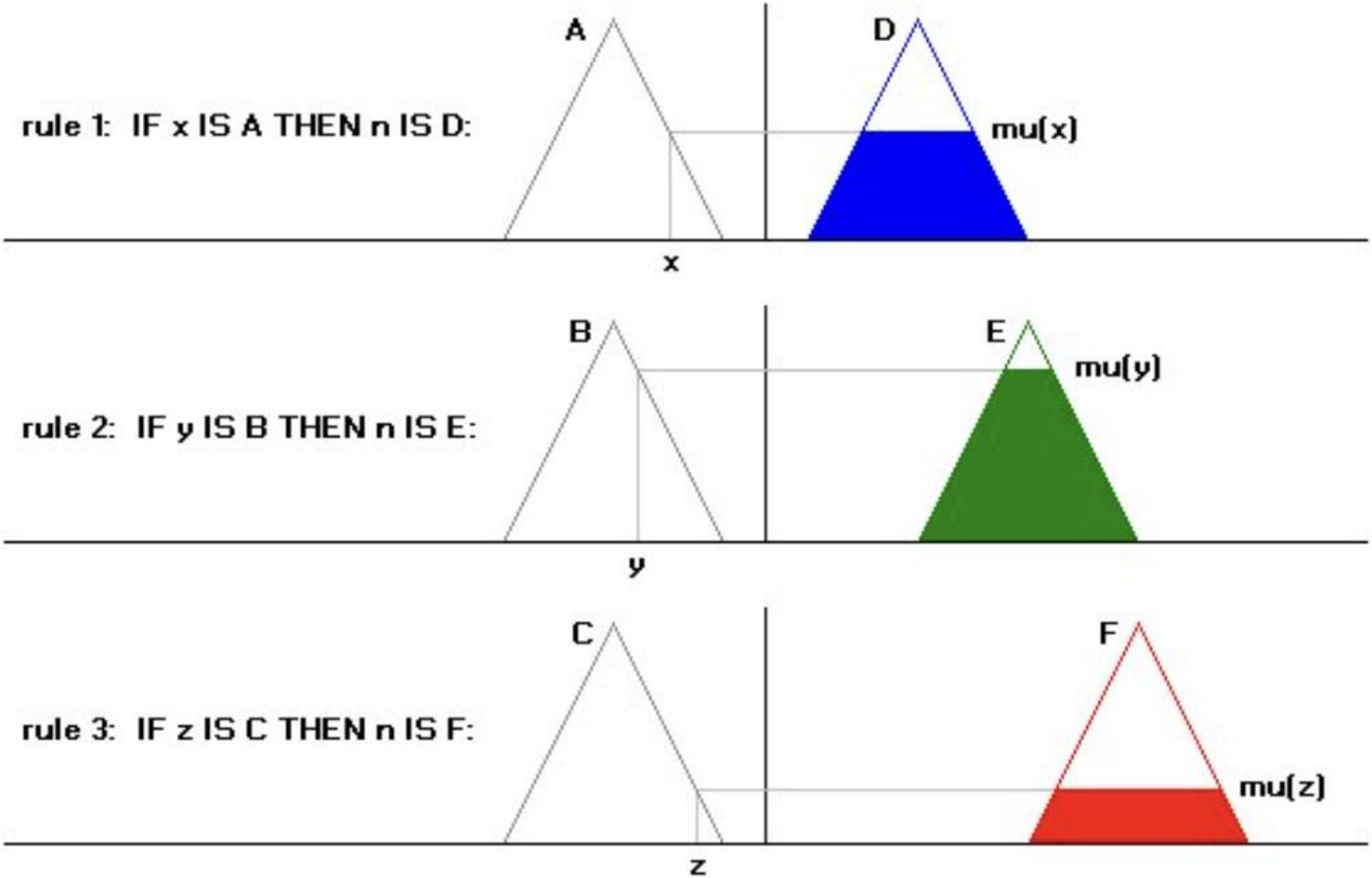
- This is a rule-based approach where membership is assigned by predefined thresholds.

- **Example:** If temperature < 20°C, it’s "Cold."

- Often used in rule-based fuzzy systems, especially for control applications.

Fuzzification

Max-Min or Threshold-Based Fuzzification:



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Fuzzification

Discretization Fuzzification:

- The input range is divided into discrete intervals, and each interval is assigned a degree of membership to a fuzzy set.

- **Example:** If temperature ranges from 0°C to 100°C, with discrete intervals (0-40, 30-80, etc.), corresponding to "Cold," "Warm," or "Hot" fuzzy sets.

- Often applied when inputs are discretized in digital systems.

Fuzzification

Method	Type of Membership Function	Advantages
Singleton	Single point	Very simple, computationally efficient
Gaussian	Smooth bell-shaped (Gaussian)	Smooth transitions, good for continuous data
Triangular/Trapezoidal	Linear, triangular, or trapezoidal	Simple, easy to implement, good for fast systems
Piecewise Linear	Custom linear	Flexible, adaptable to specific needs
Sigmoidal	S-shaped sigmoid	Good for asymmetric transitions
Max-Min	Rule-based thresholding	Efficient, especially in rule-based systems
Discretization	Interval-based	Simple and efficient for digital applications

Source: Internet

Defuzzification

- Defuzzification means the fuzzy to crisp conversion.

- The fuzzy results generate cannot be used in an application where, decision has to be taken only on crisp values.

- The defuzzification process has fuzzy variables as an input value, where the output of the defuzzification process is a crisp or numeric value.

Defuzzification

- A number of defuzzification methods are known Such as:
- Lambda-cut method
- Weighted average method
- Maxima methods
- Centroid methods

Defuzzification

Lambda-cut Method:

Lambda-cut Method is used to generate crisp value of a fuzzy set or relation. Therefore, we have:

- Lambda-cut method for fuzzy set
- Lambda-cut method for fuzzy relation
- In many literature, Lambda-cut method is also alternatively termed as Alpha-cut method.

Defuzzification

Lambda-cut Method:

- In this method a fuzzy set A is transformed into a crisp set A_λ for a given value of λ ($0 \le \lambda \le 1$)
- In other-words, $A_\lambda = \{x \mid \mu_A(x) \ge \lambda\}$
- That is, the value of Lambda-cut set A_λ is x, when the membership value corresponding to x is greater than or equal to the specified λ.
- This Lambda-cut set A_λ is also called alpha-cut set.

Defuzzification

Lambda-cut Method:

$$A = \{(x_1, 0.9), (x_2, 0.5), (x_3, 0.2), (x_4, 0.3)\}$$

Then $A_{0.6} = \{(x_1, 1), (x_2, 0), (x_3, 0), (x_4, 0)\} = \{x_1\}$

and

$$A = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.8), (x_4, 0.7)\}$$

$$A_{0.2} = \{(x_1, 0), (x_2, 1), (x_3, 1), (x_4, 1)\} = \{x_2, x_3, x_4\}$$

Defuzzification

Lambda-cut Method:

Two fuzzy sets P and Q are defined on x as follows.

$\mu(x)$	x_1	x_2	x_3	x_4	x_5
P	0.1	0.2	0.7	0.5	0.4
Q	0.9	0.6	0.3	0.2	0.8

Find the following :

(a) $P_{0.2}$, $Q_{0.3}$

(b) $(P \cup Q)_{0.6}$

(c) $(P \cup \overline{P})_{0.8}$

(d) $(P \cap Q)_{0.4}$

Source: Internet

Defuzzification

Lambda-cut Method: The Lambda-cut method for a fuzzy set can also be extended to fuzzy relation as:

Example: For a fuzzy relation R

$$R = \begin{bmatrix} 1 & 0.2 & 0.3 \\ 0.5 & 0.9 & 0.6 \\ 0.4 & 0.8 & 0.7 \end{bmatrix}$$

We are to find λ-cut relations for the following values of $\lambda = 0, 0.2, 0.9, 0.5$

$$R_0 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ and } R_{0.2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ and}$$

$$R_{0.9} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } R_{0.5} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Defuzzification Methods

A number of defuzzification methods are known. Such as:

- **Lambda-cut method**
- **Maxima methods**
- **Centroid methods**
- **Weighted average method**

Defuzzification Methods

- **Maxima Methods**

Height Method

First of Maxima (FoM)

Last of Maxima (LoM)

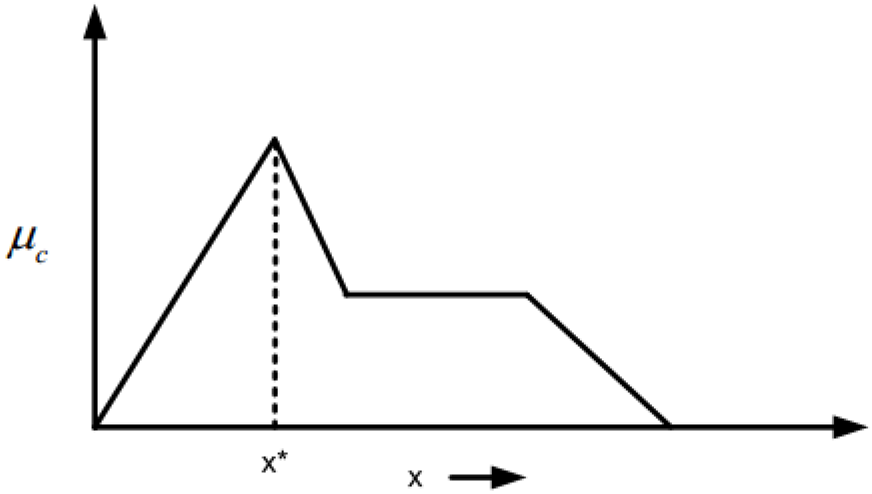
Mean of Maxima(MoM)

Defuzzification Methods

- Maxima Methods: **Height Method**

This method is defined as follows:

$$\mu_C(x^*) \geq \mu_C(x) \text{ for all } x \in X$$



- Here, x^* is the height of the output fuzzy set C.
- This method is applicable when height is unique.

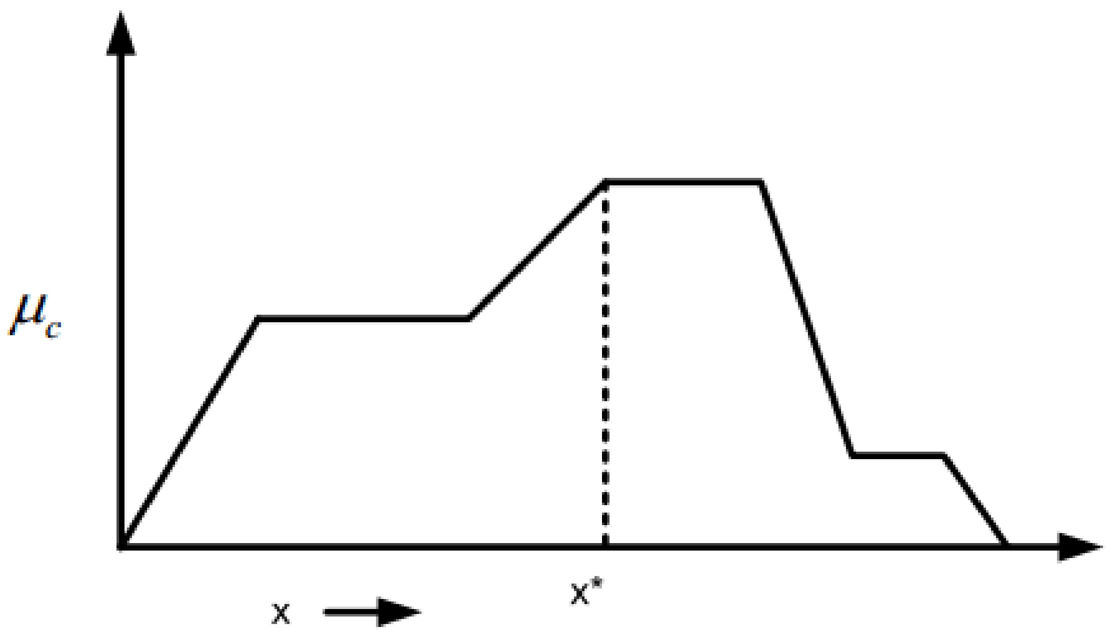
Source: Internet

Defuzzification Methods

- Maxima Methods: **First of Maxima (FoM)**

This method is defined as follows:

$$x^* = \min\{ x | C(x) = \text{max height of } C(x)\}$$



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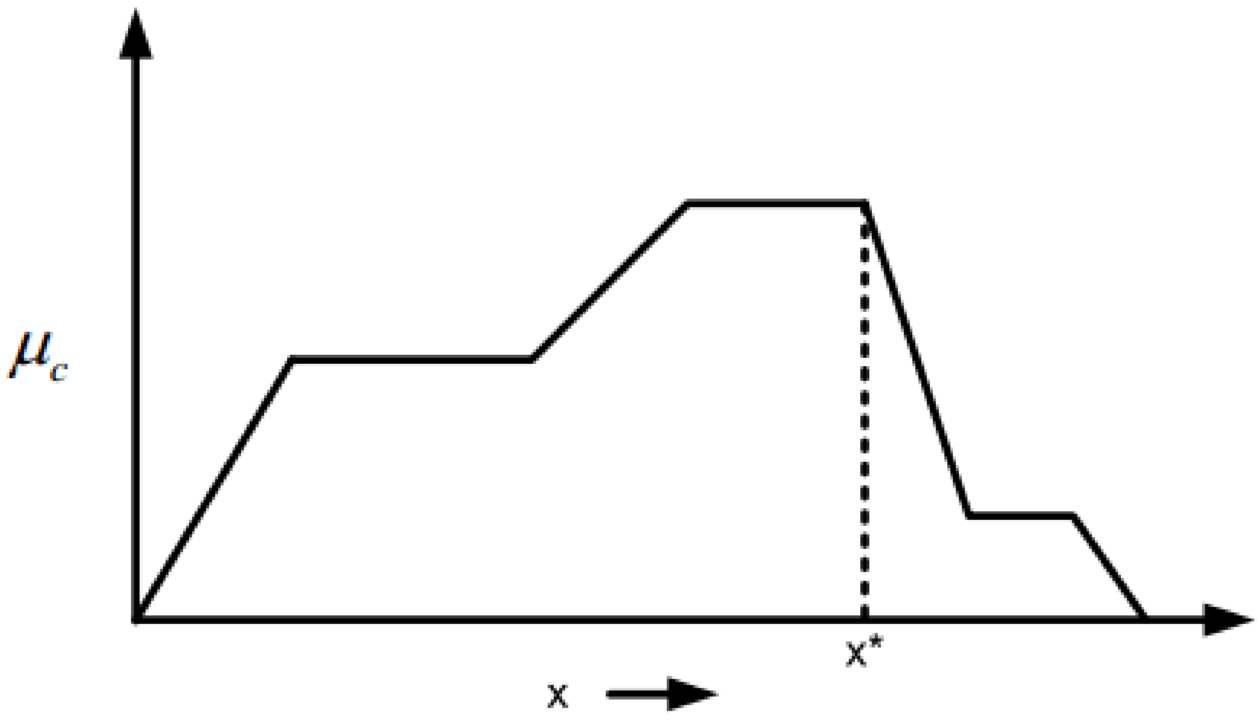
Defuzzification Methods

- Maxima Methods: **Last of Maxima (LoM)**

This method is defined as follows:

$$x^* = \max\{\, x \mid C(x) = \text{max height of } C(x)\}$$



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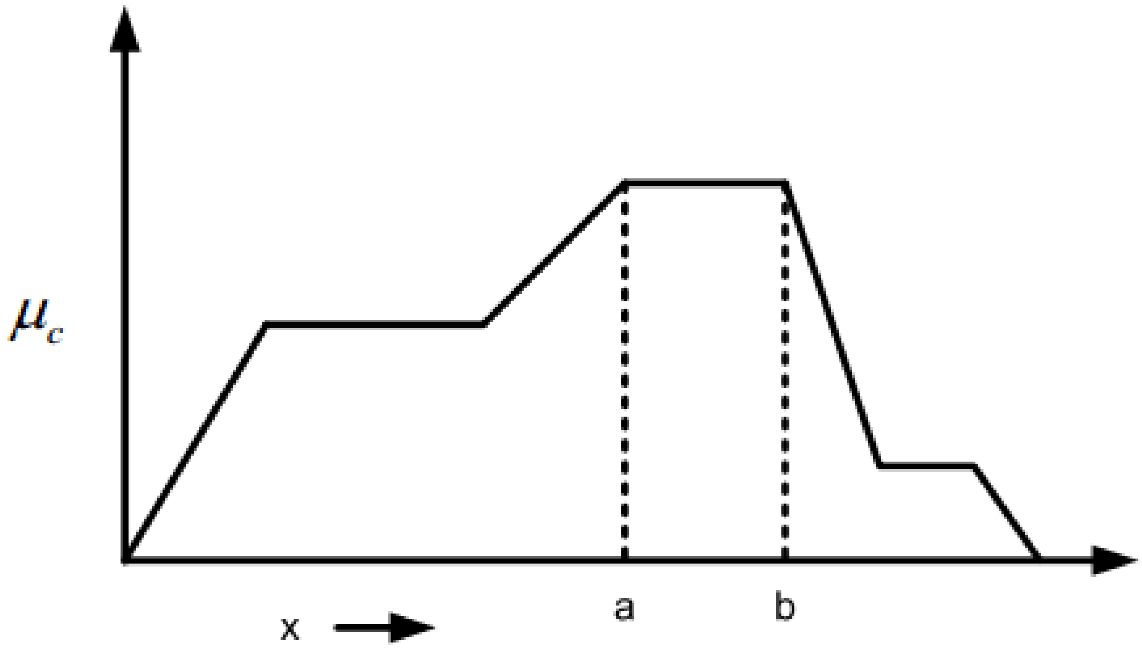
Defuzzification Methods

➢ Maxima Methods: **Mean of Maxima (MoM)**

This method is defined as follows:

$$x^* = (a+b)/2$$



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Defuzzification Methods

- **Centroid Methods**

Center of gravity method (CoG)

Center of sum method (CoS)

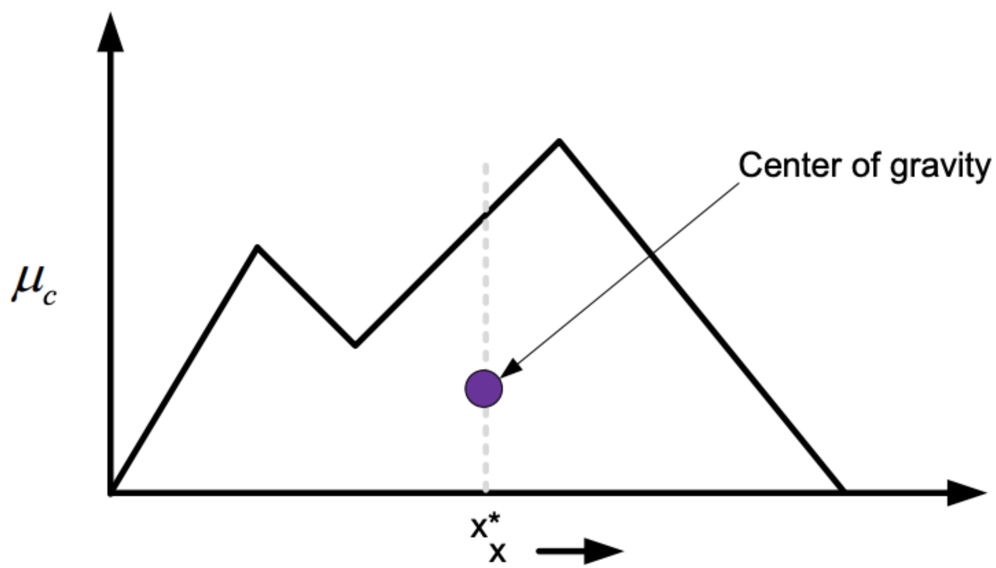
Center of area method (CoA)

Defuzzification Methods

Centroid Methods: Center of gravity method (CoG)

- The basic principle in CoG method is to find the point **x*** where a vertical line would divide the aggregate into two equal parts:
- Mathematically, the CoG can be expressed as follows:

$$x^* = \frac{\int x.\mu_C(x)dx}{\int \mu_C(x)dx}$$



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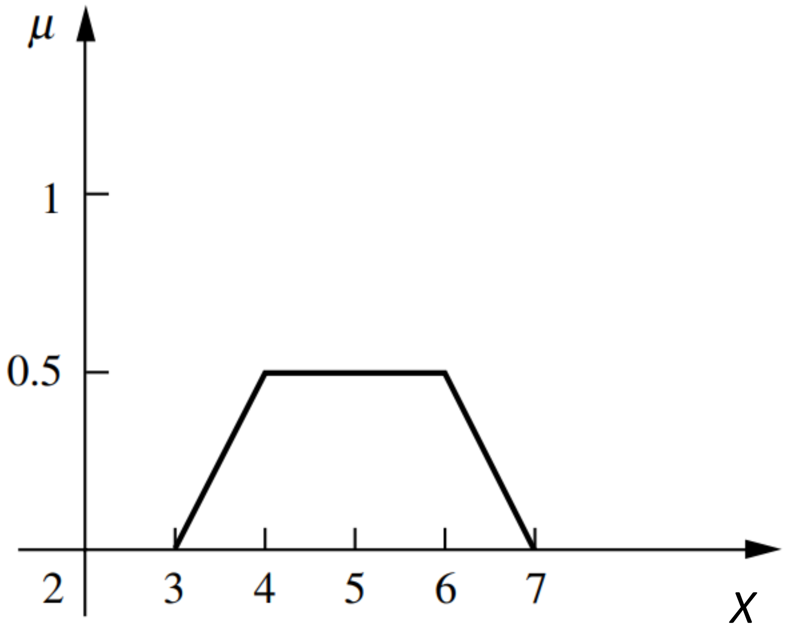
Defuzzification Methods

Centroid Methods: Center of gravity method (CoG)

- Example 1:

$$x^* = \frac{\int x.\mu_C(x)dx}{\int \mu_C(x)dx}$$



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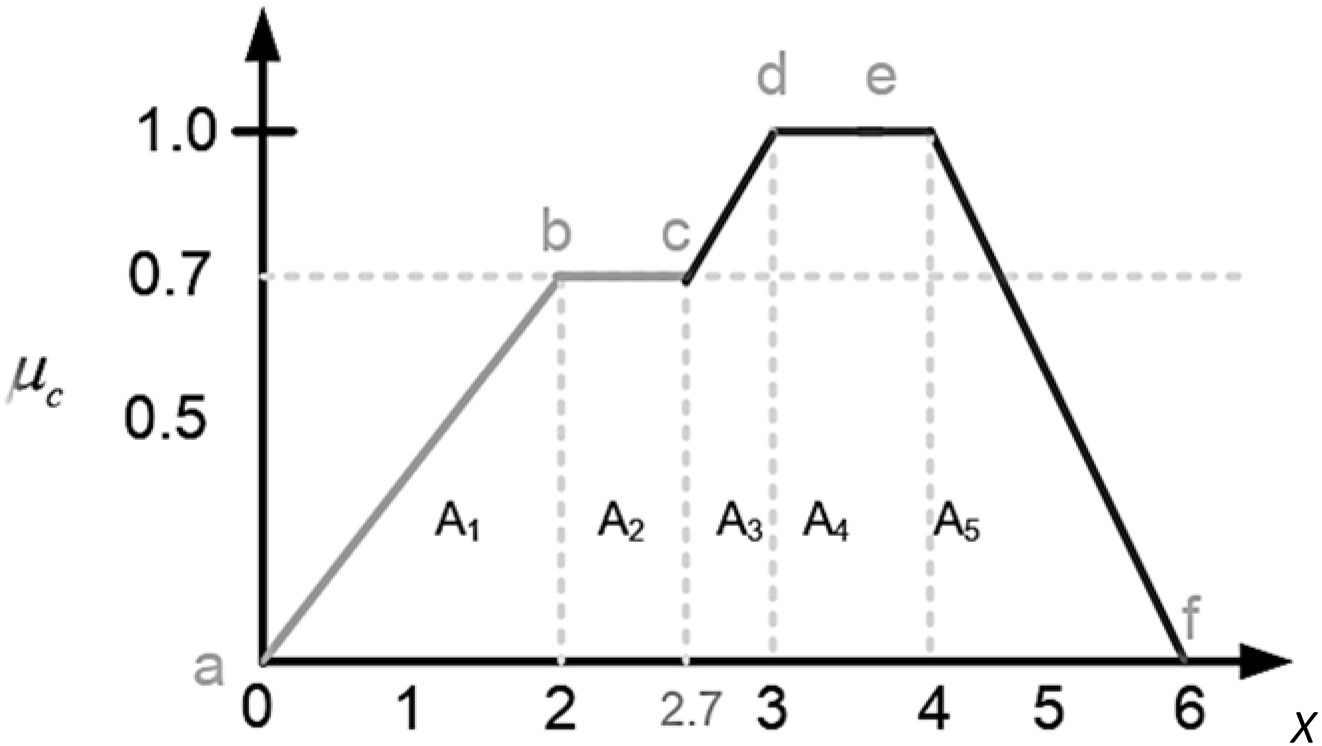
Defuzzification Methods

Centroid Methods: Center of gravity method (CoG)

- Example 2: Find $x^* = \frac{\int x.\mu_C(x)dx}{\int \mu_C(x)dx}$



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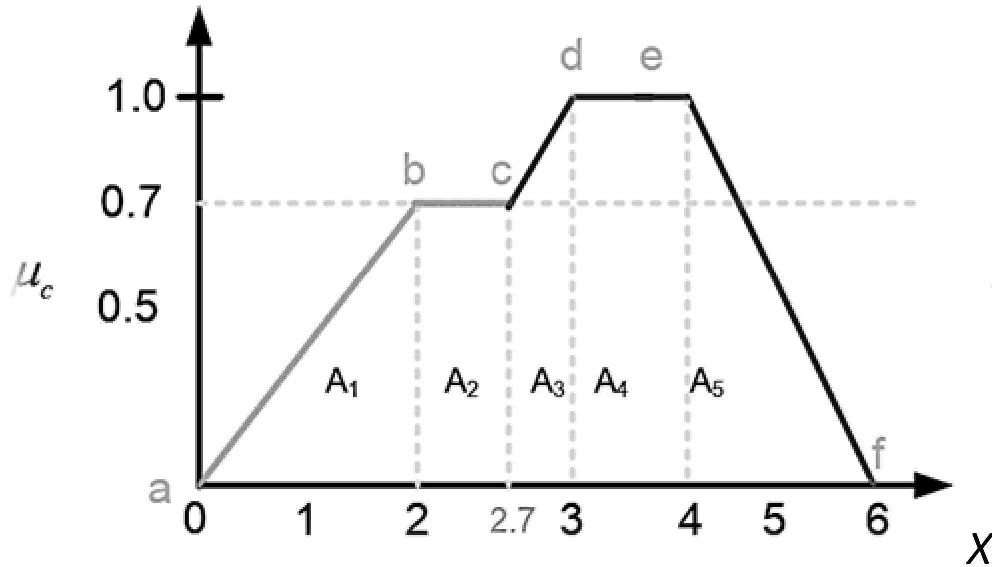
Defuzzification Methods

Centroid Methods: Center of gravity method (CoG)

- Example 2: Find



$$x^* = \frac{\int x.\mu_C(x)dx}{\int \mu_C(x)dx}$$

$$\mu_C(x) = \begin{cases} 0.35x & 0 \leq x < 2 \\ 0.7 & 2 \leq x < 2.7 \\ x - 2 & 2.7 \leq x < 3 \\ 1 & 3 \leq x < 4 \\ (-0.5x + 3) & 4 \leq x \leq 6 \end{cases}$$

Source: Internet

Defuzzification Methods

Centroid Methods: Center of gravity method (CoG)

- Example 2:

$$\mu_c(x) = \begin{cases} 0.35x & 0 \le x < 2 \\ 0.7 & 2 \le x < 2.7 \\ x - 2 & 2.7 \le x < 3 \\ 1 & 3 \le x < 4 \\ (-0.5x + 3) & 4 \le x \le 6 \end{cases}$$

Let $x^* = \frac{\int x.\mu_c(x)dx}{\int \mu_c(x)dx} = \frac{N}{D}$

$$N = \int_0^2 0.35x^2dx + \int_2^{2.7} 0.7x^2dx + \int_{2.7}^3 (x^2 - 2x)dx + \int_3^4 xdx + \int_4^6 (-0.5x^2 + 3x)dx$$

$= 10.98$

$$D = \int_0^2 0.35xdx + \int_2^{2.7} 0.7xdx + \int_{2.7}^3 (x - 2)dx + \int_3^4 dx + \int_4^6 (-0.5x + 3)dx$$

$= 3.445$

Thus, $x^* = \frac{10.98}{3.445} = 3.187$

Source: Internet

Defuzzification Methods

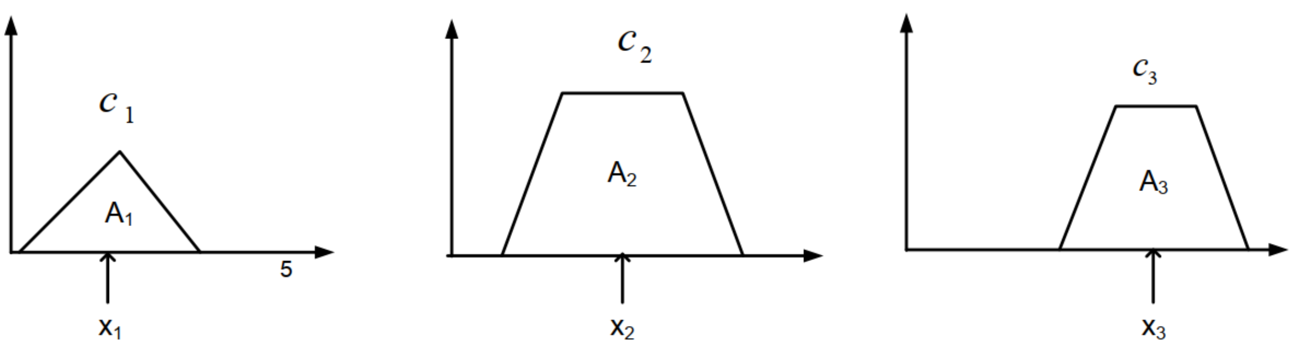
Centroid Methods: Center of Sum (CoS)

If the output fuzzy set $C = C_1 \cup C_2 \cup \ldots.C_n$, then the crisp value according to CoS is defined as

$$x^* = \frac{\sum_{i=1}^{n} x_i.A_{c_i}}{\sum_{i=1}^{n} A_{c_i}}$$

Here, A_{c_i} denotes the area of the region bounded by the fuzzy set C_i and x_i is the geometric center of the area A_{c_i}.

Graphically,



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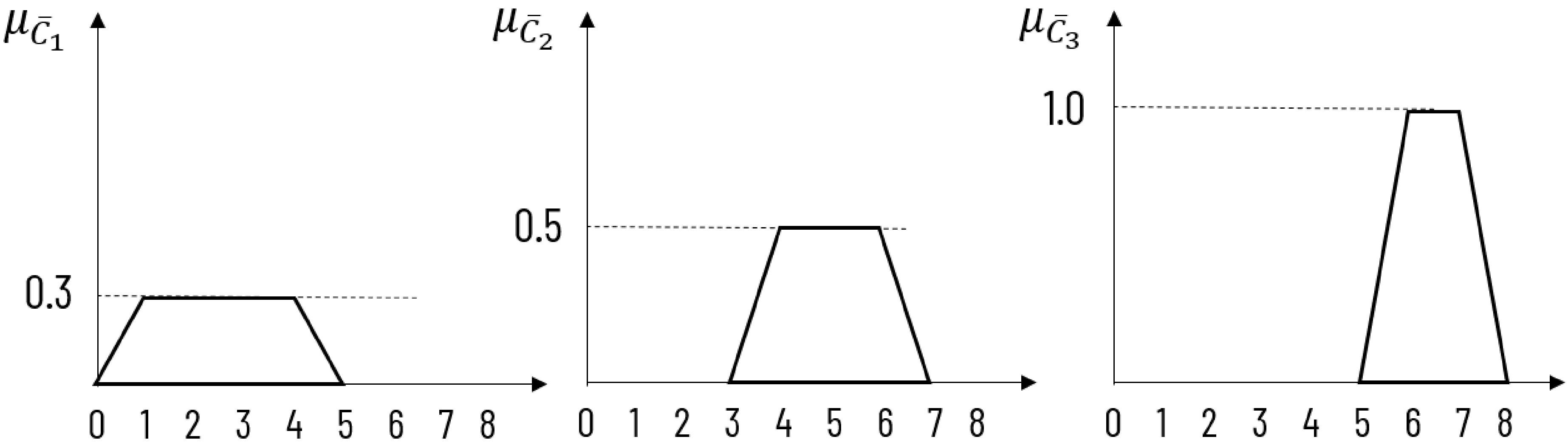
Defuzzification Methods

Centroid Methods: Center of Sum (CoS)



$$x^* = \frac{A_1 .\ x_1\ +\ A_2\ .\ x_2 +\ A_3\ .\ x_3}{A_1\ +\ A_2\ +\ A_3} = \frac{(1.2 \times 2.5) + (1.5 \times 5) + (2 \times 6.5)}{1.2 + 1.5 + 2} = 5.00$$

Source: Internet

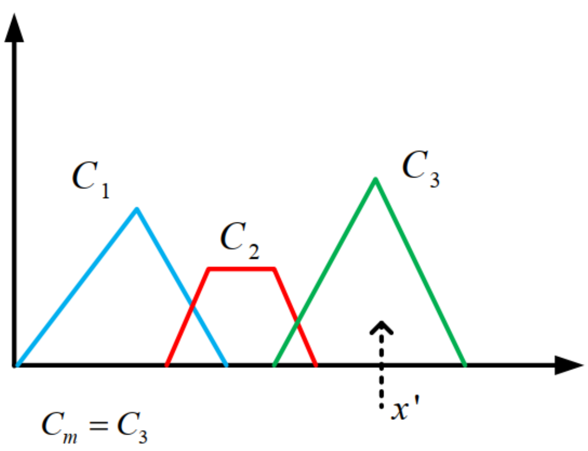
Defuzzification Methods

Centroid Methods: Center of largest Area(CoA)

If the output fuzzy set has two or more sub-regions, then the fuzzy sub-region with largest area can be used to obtain the de-fuzzified value of the output.

$$X^* = \frac{\int X \mu_{Cm}(X) d(X)}{\int \mu_{Cm}(X)\, d(X)}$$

Graphically,

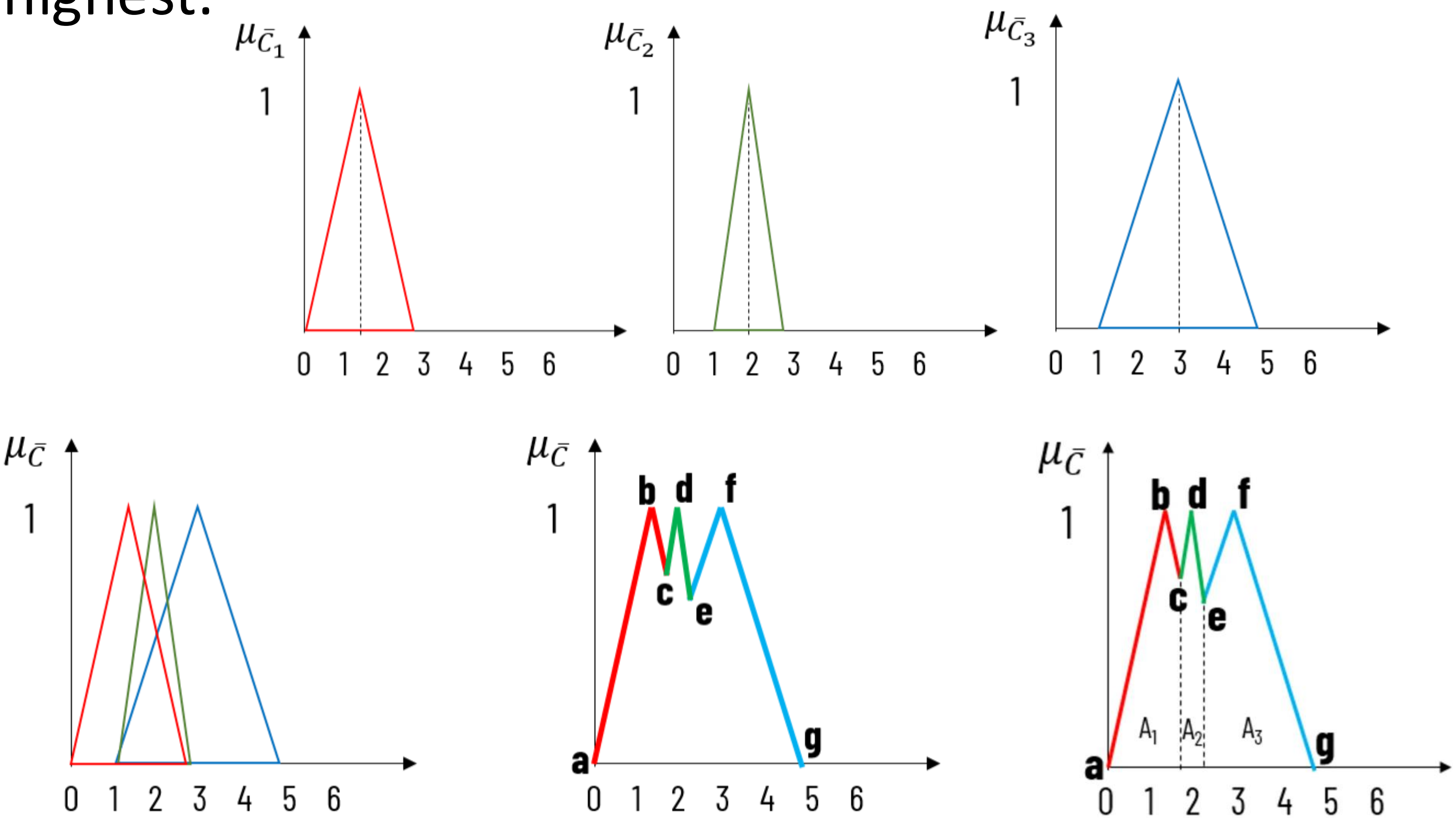


Source: Internet

Defuzzification Methods

Centroid Methods: Center of largest Area(CoA)

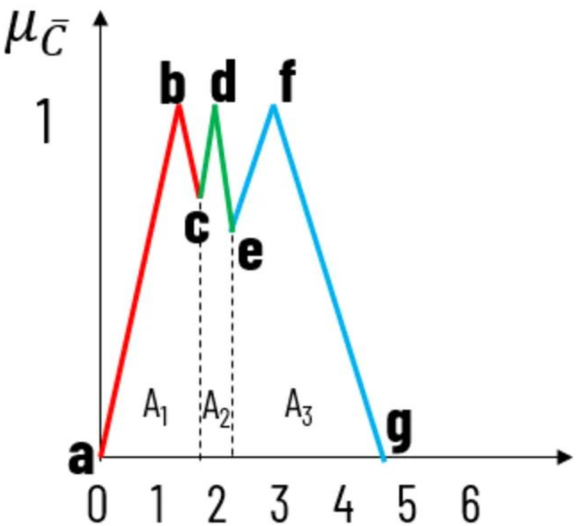
Identify all regions where the membership degree is highest.



Source: Internet

Defuzzification Methods

Centroid Methods: Center of largest Area(CoA)

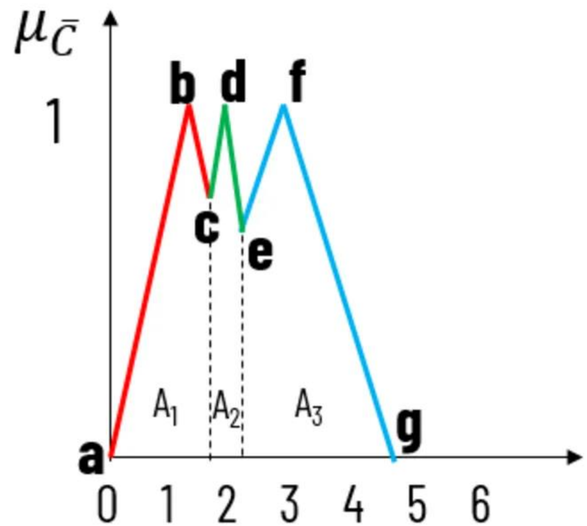


Line	Equation	Range
ab	$y = 0.67x$	[0.0, 1.5]
bc	$y = -0.67x + 2$	[1.5, 1.8]
cd	$y = x - 1$	[1.8, 2.0]
de	$y = -x + 3$	[2.0, 2.33]
ef	$y = 0.5x - 0.5$	[2.33, 3.0]
fg	$y = -0.5x + 2.5$	[3.0, 5.0]

Source: Internet

Defuzzification Methods

Centroid Methods: Center of largest Area(CoA)



$$A_1 = \int_0^{1.5}(0.67x)\,dx + \int_{1.5}^{1.8}(-0.67x+2)\,dx = 1.02$$

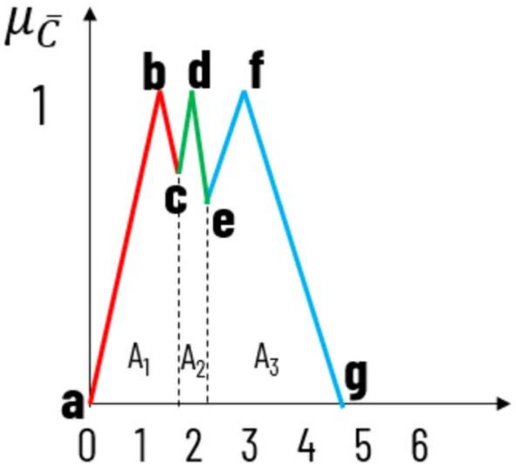
$$A_2 = \int_{1.8}^{2}(x-1)\,dx + \int_{2}^{2.33}(-x+3)\,dx = 0.46$$

$$\boldsymbol{A_3 = \int_{2.23}^{3}(0.5x-0.5)\,dx + \int_{3}^{5}(-0.5x+2.5)\,dx = 1.56}$$

Source: Internet

Defuzzification Methods

Centroid Methods: Center of largest Area(CoA)



$$x^* = \frac{\int \mu_{\bar{C}_m}(x) \, . \, x' \, dx}{\int \mu_{\bar{C}_m}(x) \, dx}$$

$$x^* = \frac{\int_{2.23}^{3}(0.5x - 0.5)x \, dx + \int_{3}^{5}(-0.5x + 2.5)x \, dx}{\int_{2.23}^{3}(0.5x - 0.5) \, dx + \int_{3}^{5}(-0.5x + 2.5) \, dx}$$

$$x^* = 3.3$$

Source: Internet

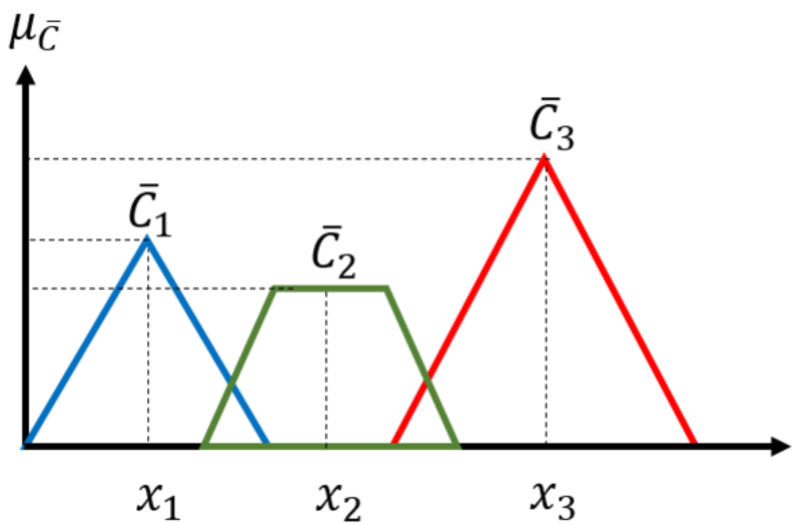
Defuzzification Methods

Weighted average method: The method can be used only for symmetrical output membership functions.

The crisp value according to this method is:

$$x^* = \frac{\sum_{i=1}^{n} \mu_{C_i}(x_i).(x_i)}{\sum_{i=1}^{n} \mu_{C_i}(x_i)}$$

where, $C_1, C_2, \ldots C_n$ are the output fuzzy sets and (x_i) is the value where middle of the fuzzy set C_i is observed.



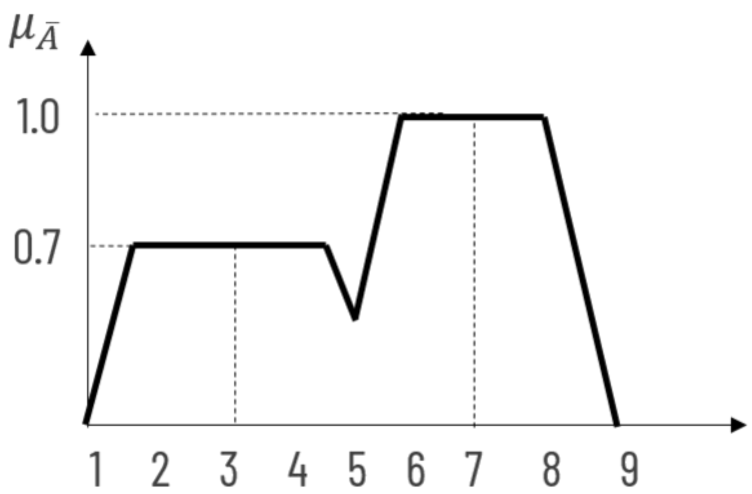
Source: Internet

Defuzzification Methods

Weighted average method: Example

$$x^* = \frac{\sum_{i=1}^{n} \mu_{C_i}(x_i).(x_i)}{\sum_{i=1}^{n} \mu_{C_i}(x_i)}$$

where, $C_1, C_2, ...C_n$ are the output fuzzy sets and (x_i) is the value where middle of the fuzzy set C_i is observed.



$x^* = ((0.7 \times 3) + (1.0 \times 7)) / (0.7 + 1.0)$

$x^* = 5.941$

Source: Internet